Does Mobility Beget Mobility?
Coworker Networks and the Sectoral Reallocation of Labor

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Job Market Paper

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Outline

Introduction

Empirical Evidence – Coworker Design

A Model of Job Choice under Coworker Influences

Policy Exercise

Conclusion

Introduction

- Labor transition tends to be sluggish. What frictions impede sectoral reallocation?
- Social networks can reduce frictions by bringing in information and skills.

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- Labor transition tends to be sluggish. What frictions impede sectoral reallocation?
- Social networks can reduce frictions by bringing in information and skills.
- We ask the following questions:
  - What is the role of coworker networks in influencing sectoral transition? We study **current coworkers' past sectors of employment**.
  - Is the relationship causal?
  - How does the coworker network affects propagation of productivity shocks?
  - How effective are policies that redirect labor across sectors, when we consider the coworker network force vs. when we do not?

Contribution: Empirical Evidence

- **Data:** German administrative linked employer-employee dataset
- **First Approach:** Coworker design with controls for confounding variables
- **Second Approach:** Quasi-experimental variation in coworker composition from worker deaths and retirements

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- **First Approach:** Coworker design with controls for confounding variables
- **Second Approach:** Quasi-experimental variation in coworker composition from worker deaths and retirements
- **Results:** Having more coworkers previously employed in a certain sector has a positive causal impact on
  - The average propensity to move to the sector
  - The elasticity of this move propensity with respect to the destination sector wage
  - The quality of the destination firm, conditional on moving

Contribution: Quantitative Model

- **Tractably incorporate** coworker network into a dynamic job choice model:
  - Workers choose jobs and sectors based on expected payoffs and costs of transitioning
  - The adjustment cost and accuracy of workers' perceived TFP both depend on the coworker network

Contribution: Quantitative Model

- **Tractably incorporate** coworker network into a dynamic job choice model:
  - Workers choose jobs and sectors based on expected payoffs and costs of transitioning
  - The adjustment cost and accuracy of workers' perceived TFP both depend on the coworker network
- **Estimate and solve** the model to examine the importance of coworker network for propagation of a productivity shock
  - Leverage the "Master Equation" approach for computation
  - Labor reallocation effect is long-lasting compared to the case without coworker networks
  - Coworker networks amplify the trade-off between aggregate wage and inequality

Related literature

- Labor movement across sectors, occupations, and firms
  - Lilien (1982), Abraham and Katz (1986), Brainard and Cutler (1993), Jovanovic and Moffitt (1990), Neal (1995), Dix-Carneiro (2014), Autor, Dorn, and Hanson (2016), Curuk and Vannoorenberghe (2017), Traiberman (2019), Humlum (2019), Jaimovich et al. (2021), Mukoyama, Takayama, and Tanaka (2024)
- Social networks in labor markets
  - Rees (1966), Granovetter (1973), Montgomery (1991), Ioannides and Loury (2004), Calvo-Armengol and Jackson (2004), Topa (2001), Dustmann et al. (2016), Glitz (2017)
- Outside options and the beliefs about outside options
  - Pissarides (2000), Postel-Vinay and Robin (2002), Moscarini (2005), Cahuc, Postel-Vinay, and Robin (2006), Bagger et al. (2014), Jarosch (2023), Stigler (1962), Conlon et al. (2018), Caldwell and Harmon (2019), Porcher (2020), Jäger et al. (2024), Bradley and Mann (2024)

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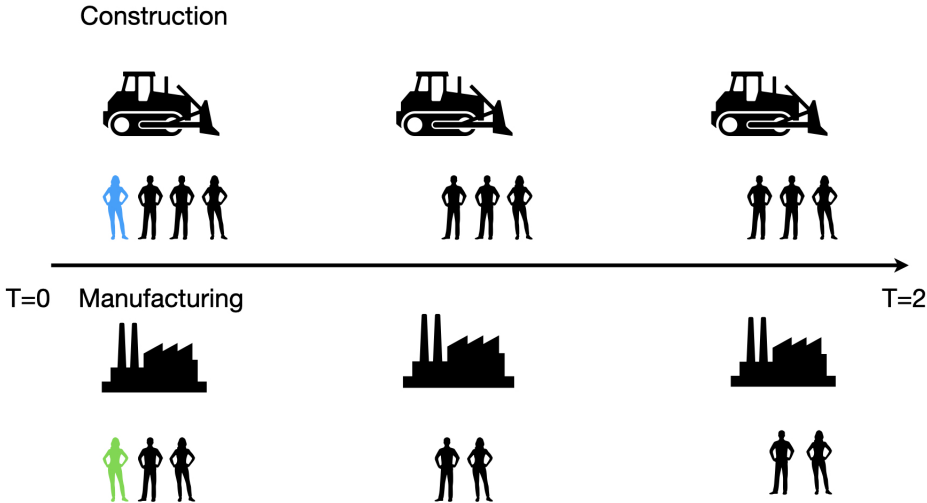
Data

- German linked employer-employee longitudinal dataset
- Includes 1.5% random sample of all establishments in Germany
- Entire employment history for the workers who are employed in the establishments sampled exact to the day

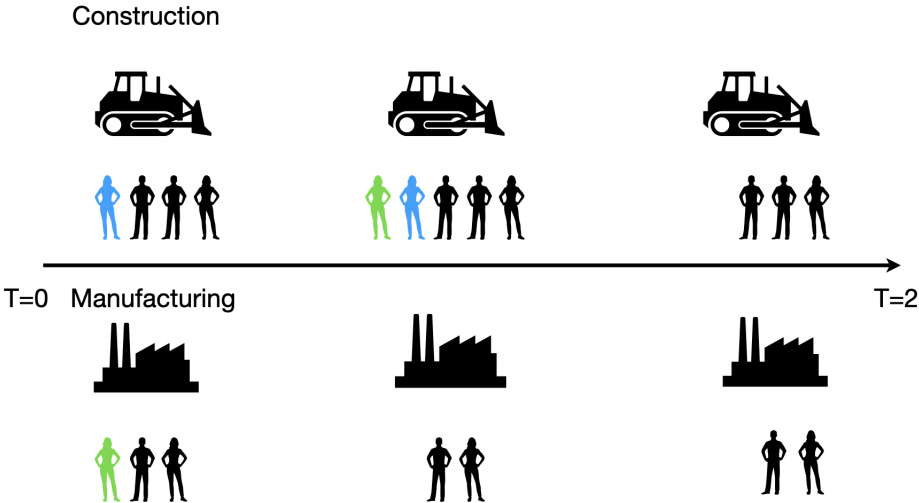
Data

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- Includes 1.5% random sample of all establishments in Germany
- Entire employment history for the workers who are employed in the establishments sampled exact to the day
- Vast majority of establishments have fewer than 100 workers
- 9 sectors and 87 occupations

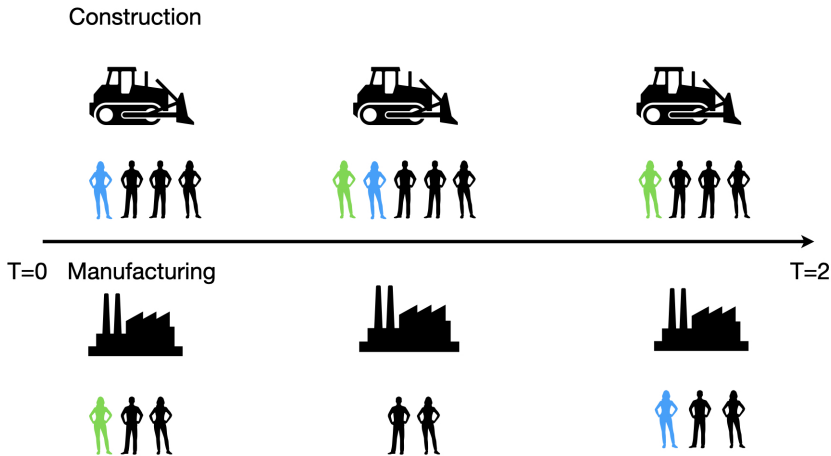
Coworker Design



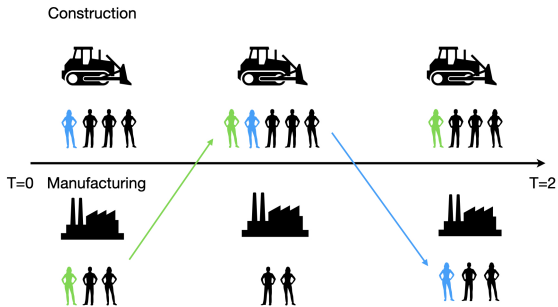
Coworker Design



Coworker Design



Implication for Labor Responses



- We emphasize bilateral flows and benefits associated with workplace interaction.

Heterogeneous Propensity to Switch Sector

$$\mathbb{1}_{pijt} = \alpha_0 + \alpha_1 \overbrace{(\ln \hat{w}_{pjt} - \ln w_{pit})}^{\text{imputed wage differential}} + \beta_1 \overbrace{(\theta_{pjt} - \frac{1}{T-1}\sum_{k \neq t} \theta_{pjk})}^{\text{current-year - non-current-year coworker}} +$$

$$\alpha_2(\theta_{pjt} - \frac{1}{T-1}\sum_{k \neq t} \theta_{pjk}) \times (\ln \hat{w}_{pjt} - \ln w_{pit}) +$$

$$\underbrace{\Gamma X_{pijt} + \zeta_{io(p,t)jt} + \xi_{l(p,t)t}}_{\text{control variables and fixed effects}} + \varepsilon_{pijt}$$

- Sample: all job switchers **who switch sectors**
- $\mathbb{1}_{pijt}$ indicates whether a worker p who is in sector i switches to sector j

Heterogeneous Propensity to Switch Sector

$$\begin{aligned}\mathbb{1}_{pijt} =& \alpha_0 + \alpha_1(\ln \hat{w}_{pjt} - \ln w_{pit}) + \beta_1(\theta_{pjt} - \frac{1}{T-1}\sum_{k\neq t}\theta_{pjk}) + \\ & \alpha_2(\theta_{pjt} - \frac{1}{T-1}\sum_{k\neq t}\theta_{pjk}) \times (\ln \hat{w}_{pjt} - \ln w_{pit}) + \\ & \Gamma X_{pijt} + \zeta_{io(p,t)jt} + \xi_{l(p,t)t} + \varepsilon_{pijt}\end{aligned}$$

- $\hat{w}_{pjt}$: the wage that p could have earned at destination sector j

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- $\hat{w}_{pjt}$: the wage that p could have earned at destination sector j
  - **Average wage** for all workers in sector j during year t who share the same occupation and location of work
  - Alternative measure using only those with the same **AKM FE** vigintile

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- θ_{pjt} ($\frac{1}{T}\sum_{t \neq k} \theta_{pjk}$) is the share of p's current-year (non-current-year) coworkers who worked in sector j for their last job. Alternative coworker specifications

Heterogeneous Propensity to Switch Sector

$$\mathbb{1}_{pijt} = \alpha_0 + \alpha_1(\ln \hat{w}_{pjt} - \ln w_{pit}) + \beta_1(\theta_{pjt} - \frac{1}{T-1}\sum_{k \neq t} \theta_{pjk}) +$$
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$$\Gamma X_{pijt} + \zeta_{io(p,t)jt} + \xi_{l(p,t)t} + \varepsilon_{pijt}$$

- $\zeta_{io(p,t)jt}$: **sector-pair** $\times$ **occupation** $\times$ **year fixed effects**
  - Absorbs **time-varying shocks** hitting each industry-pair and specific to each skill
- $\xi_{l(p,t)t}$: **location** $\times$ **year fixed effects**
- X_{pijt}: controls for individual characteristics

Alternative specification

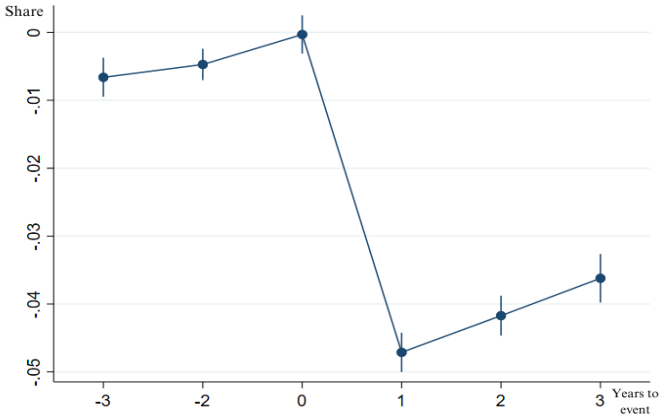
Baseline Results

	(1)	(2)	(3)	(4)	(5)
Coworker	0.121***	0.108***	0.105***	0.125***	0.116***
	(0.0009)	(0.0006)	(0.0013)	(0.0009)	(0.0009)
ln(Wage Diff)	0.005***	0.004***	0.006***	0.004***	0.005***
	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)
Coworker × ln(Wage Diff)	0.033***	0.041***	0.038***	0.028***	0.041***
	(0.0010)	(0.0008)	(0.0014)	(0.0010)	(0.0009)
Occupational-specific coworkers		✓			
Coworkers with lower wage			✓		
Controls for coworker connection				✓	
Recent-year coworker					✓
Mean of Dep. Var	0.07	0.07	0.07	0.07	0.07
R^2	0.31	0.32	0.31	0.32	0.31
Observations	151,241,259	151,241,259	151,241,259	151,159,037	151,241,259

Death and Retirement Design

- Use years relative to death/retirement as an instrument for coworker shares, in establishments that are small.
- For each deceased or retired worker-establishment pair, we match it to a placebo observation with similar worker and establishment characteristics. Matching Method
  - **Exclusion Restriction:** No additional influence of worker death or retirement on sector-transition decisions Propensity to switch
  - **Relevance:** Deaths/ retirements matter for coworker shares (firms do not immediately hire a new worker from the same origin sector)
- Sample includes **all employees** within the treated or control establishments.

Relevance of the instruments



- Coworker share from the deceased/ retired worker's last sector decreases after the death/ retirement event

Death and Retirement Instruments: First Stage

- First stage:

$$\overbrace{\theta_{ejt}}^{\text{Share in } e \text{ last employed in } j} = \alpha_0^{first} + \sum_{s=-3}^{3} \alpha_s \times \mathbb{1}_{t=d+s}$$

$$+ \sum_{t=-3}^{3} \alpha_s^{Treated} \times \mathbb{1}_{t=d+s} \times \text{Treated}_{pe} + \gamma_0^{first} \frac{1}{T-1} \sum_{k \neq t} \theta_{ejk} + \zeta_{io(p,t)jt} + \epsilon_{ejt}$$

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  - Also instrument for Share $\times$ Wage
  - Overidentified case: weighting matrix using GMM
- Second stage:

$$\mathbb{1}_{pijt} = \alpha_0 + \alpha_1(\ln \hat{w}_{pjt} - \ln w_{pit}) + \alpha_2 \widehat{(\text{Share} \times \text{Wage})}_{pijt} + \beta_1 \hat{\theta}_{pjt}$$

$$+ \gamma_0^{first} \frac{1}{T-1} \sum_{k \neq t} \theta_{ejk} + \zeta_{io(p,t)jt} + \varepsilon_{pijt}$$

Death and Retirement Analysis Results

	(1)	(2)	(3)
Coworker	0.049***	0.021**	0.137***
	(0.0047)	(0.0106)	(0.0122)
ln(Wage Diff)	0.011***	0.009***	0.005
	(0.0013)	(0.0029)	(0.0035)
Coworker × ln(Wage Diff)	0.044***	0.040***	0.062***
	(0.0057)	(0.0137)	(0.0147)
Sample	All	Est. with Deaths	New Hires
Mean of Dep. Var	0.061	0.066	0.137
Observations	5,334,545	411,399	697,496

Interpretation of Results

- Average propensity to move to a sector increases by 12.1 p.p. among all **job-switchers** if the difference between current-year and non-current year coworker share changes from 0 to 1 (4.9 p.p. among **all workers**)
- A 50% wage increase corresponds to an additional 1.7 p.p. likelihood of moving among **job-switchers** (2.2 p.p. among **all workers**)

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  - Workers also move to establishments with higher firm AKM FEs and exhibit longer tenure at the new job Firm Choice

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Firms

- Firms: $N \in \mathcal{N}$ sectors, $M_n \in \mathcal{M}_n$ firms within sector n. Firms within the same sector produce perfectly substitutable goods
- Production: A firm jn chooses labor to maximize profit under the production function with decreasing returns to scale:

$$Y_t^{jn} = \underbrace{B^{jn}}_{\text{firm-level TFP}} \underbrace{Z_t^n}_{\text{sector-level TFP}} (L_t^{jn})^{1-\gamma}, \ \gamma > 0$$

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- TFP: Fixed firm-specific component and **time-varying sectoral TFP shocks**

$$d \log \frac{Z_t^n}{Z^n} = -\eta \log \frac{Z_t^n}{Z^n} dt + \epsilon^n \delta^n dW_t; \ z_t^n = \frac{1}{\epsilon^n \delta^n} \log \frac{Z_t^n}{Z^n}$$

Workers

- Labor supply:
  - Each worker supplies one unit of labor to a firm
  - At rate ϕ they can switch firms, incurring an **adjustment cost** of $\kappa_t^{jn,ik}$
  - Each choice involves a **taste shock**, which is **correlated by** roughly $1-\zeta$ for firms within the same sector and uncorrelated for firms across sectors

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  - **Flows**: $\mu_t^{jn,ik}$ is the fraction of workers in jn that move to ik; $l_t^{jn,ik}$ is the fraction of workers in ik that come from jn

$$l_t^{n,ik} = \sum_{r \in \mathcal{M}_n} \frac{\mu_t^{rn,ik} L_t^{rn}}{L_t^{ik}}, \quad l_t^{jn,ik} = \frac{\mu_t^{jn,ik} L_t^{jn}}{L_t^{ik}}$$

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- Consumption: Workers are hands-to-mouth and consume by maximizing a Cobb-Douglas aggregate of sectoral goods:

$$c_t^{jn} = \Pi_{k=0}^{N} (c_t^{jn,k})^{\gamma^k} = w_t^{jn} \Big/ [\Pi_{k=1}^{N} (p_t^k / \gamma^k)^{\gamma^k}]$$

Coworker Influences

- Perceived Wage:
  - Workers in the same firm *jn* share **identical beliefs** about the firm- and sector-specific components of TFP for firm *ik*, denoted by $\hat{z}_t^{jn,k}$ and $\hat{b}_t^{jn,ik}$.
  - The **accuracy** of their beliefs **increases** with the share of current coworkers that were previously employed in the potential destination sector of employment.

$$\hat{z}_t^{jn,k} = \beta_0 l_t^{k,jn} \log(Z_t^k) + (1 - \beta_0 l_t^{k,jn})\bar{z}$$

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- Adjustment Cost:
  - Adjustment cost from firm *jn* to *ik* consists of a **sector** component and a **firm** component, both **decreasing** with the size of the coworker network

$$\kappa_t^{jn,ik} = \underbrace{(1 - \alpha_0 l_t^{k,jn})\bar{\kappa}_t^{n,k}}_{\text{sector-level cost}} + \underbrace{(1 - \alpha_1 l_t^{ik,jn})\bar{\kappa}}_{\text{firm-level cost}}$$

Job Flows under Coworker Influence

- Sector-level Flows:

$$\mu_t^{jn,k} = \frac{\exp\left(\zeta \overbrace{S_t^{jn,k}}^{\text{inclusive value}} + \overbrace{\frac{1-\beta_0 l_t^{k,jn}}{\nu(\rho+\phi)}(\bar{z} - \log(Z_t^k))}^{\text{elasticity to sector TFP}} - \overbrace{\frac{1}{\nu}(1-\alpha_0 l_t^{k,jn})\bar{\kappa}_t^{n,k}}^{\text{sector-level adjustment cost}}\right)}{\sum_{h\in\mathcal{N}} \exp\left(\zeta S_t^{jn,h} + \frac{1-\beta_0 l_t^{h,jn}}{\nu(\rho+\phi)}(\bar{z} - \log(Z_t^h)) - \frac{1}{\nu}(1-\alpha_0 l_t^{h,jn})\bar{\kappa}_t^{n,h}\right)}$$

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- Firm-level Flows:

$$\mu_t^{jn,ik|k} = \frac{\exp\left(V_t^{ik} - \overbrace{\frac{1-\beta_1 l_t^{k,jn}}{\rho+\phi}\log(B^{ik})}^{\text{elasticity to firm TFP}} - \overbrace{(1-\alpha_1 l_t^{ik,jn})\bar{\kappa}}^{\text{firm-level adjustment cost}}\right)^{\frac{1}{\nu\zeta}}}{\sum_{r\in\mathcal{M}_k} \exp\left(V_t^{rk} - \frac{1-\beta_1 l_t^{k,jn}}{\rho+\phi}\log(B^{rk}) - (1-\alpha_1 l_t^{rk,jn})\bar{\kappa}\right)^{\frac{1}{\nu\zeta}}}$$

Estimation of Labor Demand Shocks and Coworker Parameters

1. Estimate **productivity parameters** from relationships between Z^n_{t+1} and Z^n_t
   - Recover changes in labor demand similar to the one experienced in COVID using productivity changes by sector
   - Positively impacted sectors include Information, Utilities, and Agriculture; Negatively impacted sectors include Services, Manufacturing COVID shock

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3. **Revealed preference approach** identifies the job-switching elasticity, ν Job-switching elasticity

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3. **Revealed preference approach** identifies the job-switching elasticity, ν Job-switching elasticity
4. **Cross-sector flows** identify the sector-level coworker network, α_0 and β_0, and the baseline sector-level transition costs, $\bar{\kappa}^{n,k}$ Sector-level Network

Parameter Values

Firm-level Network

$$\log \frac{\overbrace{\mu_t^{jn,rn|n}}^{j\to r}}{\mu_t^{jn,jn|n}} + \log \frac{\overbrace{\mu_t^{rn,jn|n}}^{r\to j}}{\mu_t^{rn,rn|n}} + \frac{1}{\nu\zeta(\rho+\phi)}(\log B^{rn} + \log B^{jn})$$
$$= \frac{\beta_1}{\nu\zeta(\rho+\phi)}(l_t^{n,jn}\log B^{rn} + l_t^{n,rn}\log B^{jn}) + \frac{\alpha_1}{\nu\zeta}(l_t^{rn,jn} + l_t^{jn,rn})\bar{\kappa} - \frac{2}{\nu\zeta}\bar{\kappa}$$

- $l_t^{n,jn}\log B^{rn} + l_t^{n,rn}\log B^{jn}$: sum of coworker network weighted firm-level TFP
- $(l_t^{rn,jn} + l_t^{jn,rn})\bar{\kappa}$: sum of coworker network weighted transition costs

back

The Impact of Coworker Networks

- In our framework

$$\mu_t^{ik,jn} = \Omega^{ik,jn}(\mu_t, L_t, z_t, V_t)$$

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$$\mu_t^{ik,jn} = \Omega^{ik,jn}(\mu_t, L_t, z_t, V_t)$$

- Total differentiation yields:

$$d\mu_t^{ik,jn} = \sum_{m,s,x,q} \overbrace{\frac{\partial \Omega^{ik,jn}}{\partial \mu_t^{ms,xq}}}^{T_{ikjn,msxq}} d\mu_t^{ms,xq} + \sum_{dh} \overbrace{\frac{\partial \Omega^{ik,jn}}{\partial L_t^{dh}}}^{R_{ikjn,dh}} dL_t^{dh} + \sum_{dh} \overbrace{\frac{\partial \Omega^{ik,jn}}{\partial V_t^{dh}}}^{G_{ikjn,dh}} dV_t^{dh} + \overbrace{\frac{\partial \Omega^{ik,jn}}{\partial z_t}}^{Q_{ikjn}} dz_t$$

$$d\mu_t = \underbrace{(Id - T)^{-1}}_{\text{"Leontief Inverse" for the labor network}} (RdL + GdV + Qdz)$$

The Impact of Coworker Networks

- In our framework

$$\mu_t^{ik,jn} = \Omega^{ik,jn}(\mu_t, L_t, z_t, V_t)$$

- Total differentiation yields:

$$d\mu_t^{ik,jn} = \sum_{m,s,x,q} \overbrace{\frac{\partial \Omega^{ik,jn}}{\partial \mu_t^{ms,xq}}}^{T_{ikjn,msxq}} d\mu_t^{ms,xq} + \sum_{dh} \overbrace{\frac{\partial \Omega^{ik,jn}}{\partial L_t^{dh}}}^{R_{ikjn,dh}} dL_t^{dh} + \sum_{dh} \overbrace{\frac{\partial \Omega^{ik,jn}}{\partial V_t^{dh}}}^{G_{ikjn,dh}} dV_t^{dh} + \overbrace{\frac{\partial \Omega^{ik,jn}}{\partial z_t}}^{Q_{ikjn}} dz_t$$

$$d\mu_t = \underbrace{(Id - T)^{-1}}_{\text{"Leontief Inverse" for the labor network}} (RdL + GdV + Qdz)$$

- $(Id - T)^{-1}$ has norm and eigenvalue greater than 1 $\Rightarrow$ coworker network amplifies the effect of changes in values obtainable at each sector/ firm

Outline

Introduction

Empirical Evidence – Coworker Design

A Model of Job Choice under Coworker Influences

Policy Exercise

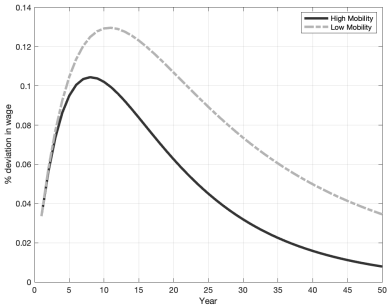
Conclusion

Coworker networks amplify labor and wage responses to shock

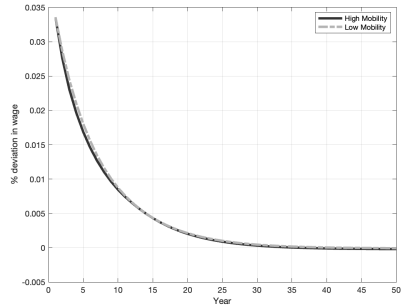
- Solve **impulse responses** of labor and wages using the "Master Equation" approach, treating labor distribution as a state variable and calculating first-order approximation around the steady state

Coworker networks amplify labor and wage responses to shock

- Solve **impulse responses** of labor and wages using the "Master Equation" approach, treating labor distribution as a state variable and calculating first-order approximation around the steady state



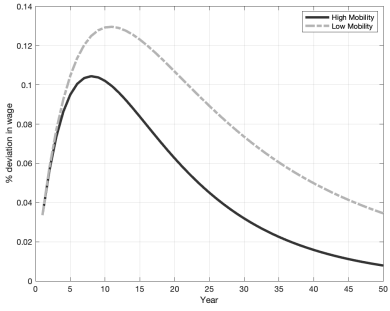
(a) Coworker Network



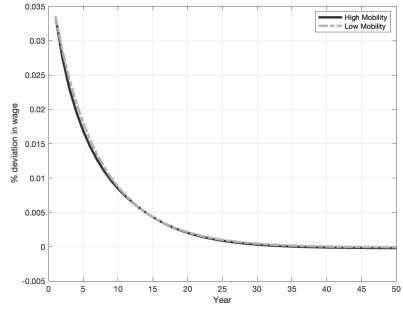
(b) No Network ($\alpha_1, \beta_1, \alpha_0, \beta_0 = 0$)

Labor Distribution

Coworker networks highlight the difference between high and low mobility

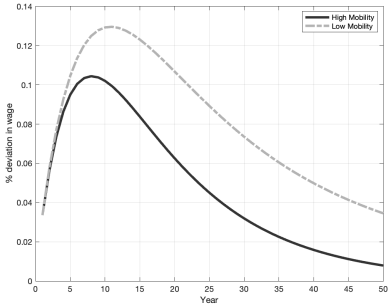


(a) Coworker Network

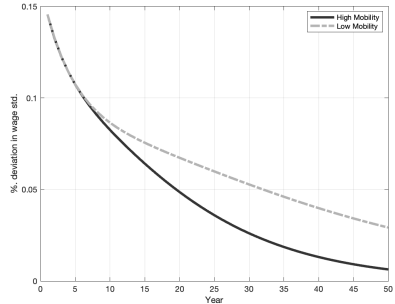


(b) No Network ($\alpha_1, \beta_1, \alpha_0, \beta_0 = 0$)

Coworker networks emphasize aggregate wage and inequality tradeoff



(a) Aggregate Wage under Coworker Network



(b) Standard Deviation under Coworker Network

- Aggregate wage increases in the long run with lower mobility
- Standard deviation in wage is higher when mobility is low

Welfare

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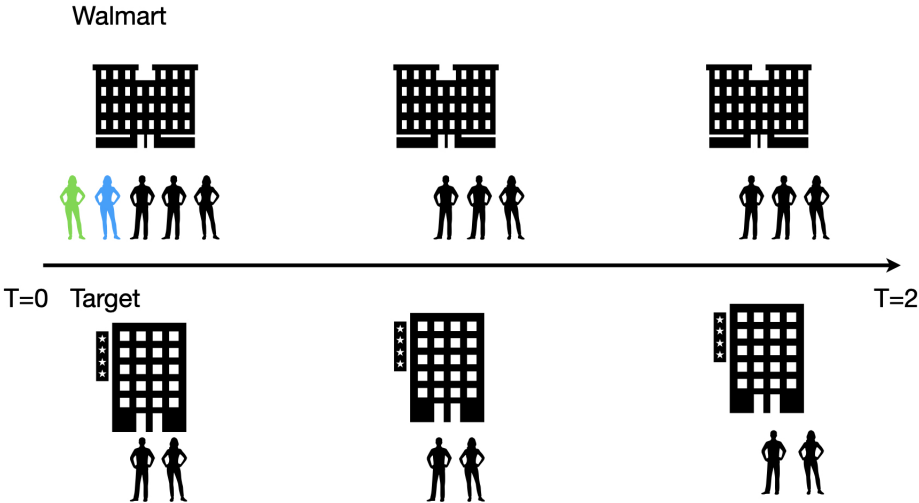
Conclusion

- **Causal relationship** between one's previous sectoral composition of current coworkers and her job-switching decisions
  - Average **propensity** to move to a sector
  - **Elasticity** of moving decision with respect to wage
  - **Quality** of firm within a sector
- **Tractable model** that incorporates the influence from the coworker network
  - Labor allocation effect is **long-lasting**
  - Emphasis of the trade-off between **aggregate welfare** and **inequality**
- Our framework opens the door to future work
  - Other types of coworker or social networks; Other types of shocks (e.g. trade, automation)
  - Firm responses under the existence of coworker networks
  - Normative implications for job search, growth, and industrial policies
  - Labor network and production network

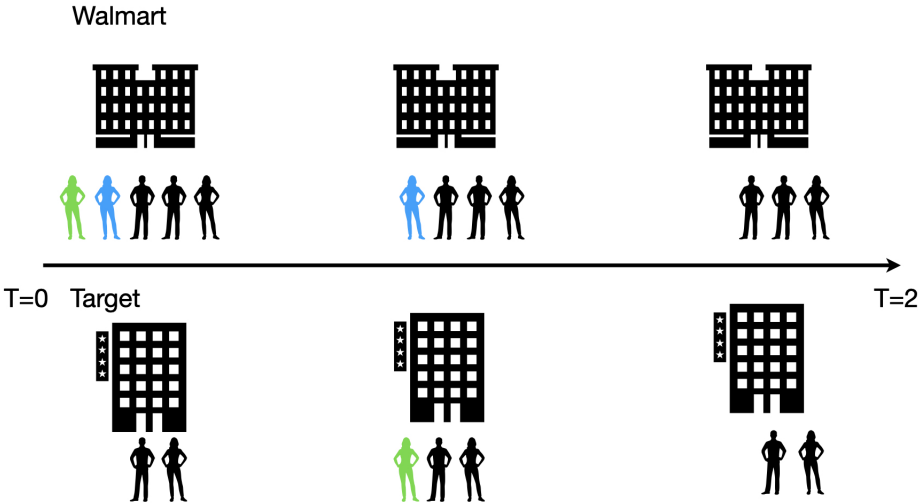
Thank you!

yiningmo@fas.harvard.edu

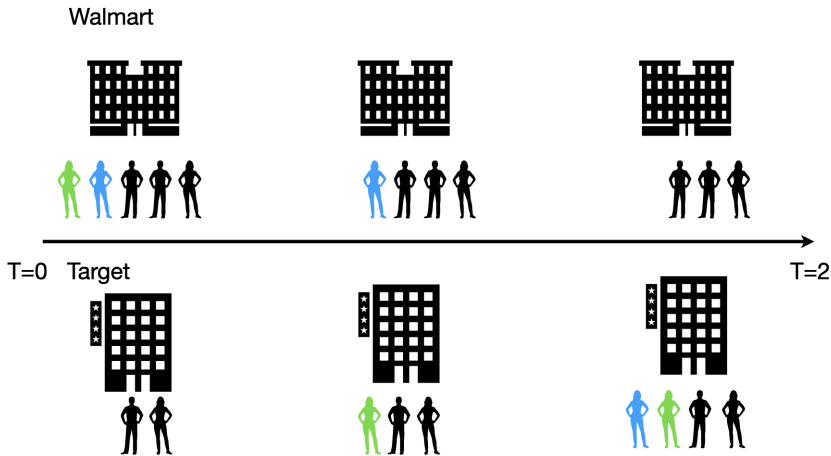
Coworker Design in Previous Work



Coworker Design in Previous Work



Coworker Design in Previous Work



back

Alternative Coworker Definitions

1. Coworkers in the **same occupation**
2. Coworkers with **below-average wage**
3. Using only **recent years** or **previous year** for measures of non-current-year coworkers
4. Excluding as coworkers those who transition to p's destination firm before p does

back

Results: Alternative coworker controls

$$\mathbb{1}_{pijt} = \alpha_0 + \Gamma X_{pijt} + \zeta_{io(p,t)jt} + \xi_{l(p,t)t} + \alpha_1(\ln \hat{w}_{pjt} - \ln w_{pit}) + \alpha_2 \theta_{pjt} \times (\ln \hat{w}_{pjt} - \ln w_{pit}) + \alpha_3 \frac{1}{T-1} \sum_{k \neq t} \theta_{pjk} \times (\ln \hat{w}_{pjt} - \ln w_{pit}) + \beta_1 \theta_{pjt} + \beta_2 \frac{1}{T-1} \sum_{k \neq t} \theta_{pjk} + \varepsilon_{pijt}$$

	(1)	(2)
Coworker	0.117***	0.147***
	(0.0009)	(0.0011)
ln(Wage Diff)	0.003***	0.003***
	(0.0001)	(0.0001)
Coworker × ln(Wage Diff)	0.038***	0.021***
	(0.0009)	(0.0006)
Separate controls	✓	
One-year lag		✓
Mean of Dep. Var	0.06	0.06
R^2	0.32	0.32
Observations	196,592,086	159,469,014

back

Results: Gender and Nationality

	Male	Female	Foreign	German
Coworker	0.123***	0.109***	0.114***	0.121***
	(0.0010)	(0.0013)	(0.0017)	(0.0009)
ln(Wage Diff)	0.004***	0.006***	0.002***	0.006***
	(0.0001)	(0.0001)	(0.0001)	(0.0001)
Coworker × ln(Wage Diff)	0.036***	0.036***	0.034***	0.033***
	(0.0012)	(0.0014)	(0.0023)	(0.0010)
Mean of Dep. Var	0.073	0.074	0.073	0.074
R^2	0.32	0.31	0.40	0.31
Observations	94,210,240	56,977,554	14,776,011	136,408,129

back

Results: Age

	(1)	(2)	(3)	(4)	(5)
Coworker	0.079***	0.118***	0.133***	0.149***	0.171***
	(0.0008)	(0.0011)	(0.003)	(0.0016)	(0.0022)
ln(Wage Diff)	0.004***	0.005***	0.005***	0.005***	0.005***
	(0.0001)	(0.0001)	(0.0001)	(0.0002)	(0.0001)
Coworker $\times$ ln(Wage Diff)	0.018***	0.036***	0.039***	0.034***	0.042***
	(0.0014)	(0.0017)	(0.0016)	(0.0020)	(0.0024)
Mean of Dep. Var	0.083	0.075	0.070	0.067	0.062
R^2	0.37	0.31	0.29	0.28	0.26
Observations	47,061,425	29,742,688	33,281,554	22,102,115	18,843,929

back

Results: Tenure

	(1)	(2)	(3)	(4)	(5)
Coworker	0.079***	0.121***	0.138***	0.154***	0.177***
	(0.0008)	(0.0011)	(0.0012)	(0.0015)	(0.0021)
ln(Wage Diff)	0.003***	0.005***	0.005***	0.005***	0.005***
	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)
Coworker $\times$ ln(Wage Diff)	0.014***	0.032***	0.033***	0.029***	0.032***
	(0.0013)	(0.0017)	(0.0016)	(0.0019)	(0.0024)
Mean of Dep. Var	0.083	0.075	0.070	0.067	0.062
R^2	0.37	0.31	0.29	0.28	0.26
Observations	47,055,187	29,745,723	33,271,123	22,094,320	18,838,162

back

Results: Alternative Samples

	(1)	(2)	(3)	(4)	(5)
Coworker	0.114***	0.113***	0.149***	0.071***	0.091***
	(0.0010)	(0.0008)	(0.0011)	(0.0019)	(0.0002)
ln(Wage Diff)	0.006***	0.005***	0.006***	0.003***	0.0001***
	(0.0001)	(0.0001)	(0.0002)	(0.0001)	(0.0000)
Coworker × ln(Wage Diff)	0.039***	0.040***	0.022***	0.030***	0.044***
	(0.0011)	(0.0009)	(0.0009)	(0.0019)	(0.0002)
Exclude Geography Switching	✓				
Exclude Top-coded Wages		✓			
Exclude Zero-valued Flows			✓		
Exclude Switching to Past Sector				✓	
Establishment-level Analysis					✓
Mean of Dep. Var	0.072	0.074	0.130	0.057	0.131
R^2	0.32	0.32	0.32	0.28	0.15
Observations	124,941,253	145,494,761	39,571,954	149,917,307	47,966,424

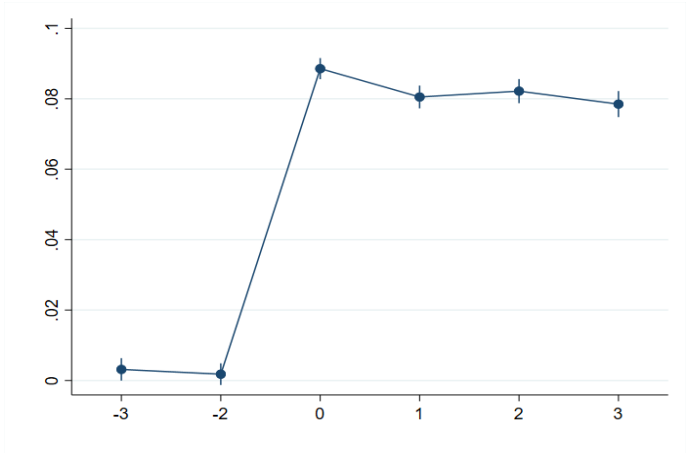
back

Matching Method to find Control Group

- For each deceased or retired worker-establishment pair, match on the following characteristics:
  - **Worker Characteristics**: nationality (Germany vs. foreign), gender (Female vs. Male), skill level (skilled vs. non-skilled), education level (secondary/intermediate school vs. upper secondary school vs. college), full-time status (full-time vs. part-time), age (quartile), wage(quartile), tenure (quartile), last sector of employment, time since switching to the current job (more than vs. less than one year) at time d;
  - **Establishment Characteristics**: size (quintile), firm fixed effect (quartile), share of workers in industry j (above vs. below 50%) at time d.
- For multiple matches, select the placebo observation with the highest propensity score.
- Able to find matches for 75% of the actual deaths.

back

Propensity to switch job increases after coworker exit



- Workers are more likely to leave establishment after worker death or retirement

back

Coworker Approach: Firm Choice

$$y_{pijt} = \alpha_0 + \alpha_1(\theta_{pjt} - \frac{1}{T-1}\sum_{k \neq t}\theta_{pjk}) + \alpha_2 \frac{1}{T-1}\sum_{k \neq t}\theta_{pjk} + \zeta_{ijt} + \xi_{l(p,t)t} + \Gamma X_{pijt} + \varepsilon_{pijt}$$

- Does having more coworkers coming from a certain industry makes job switchers more likely to move to "better quality" firms within a sector?

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- Sample includes all individuals that transition into sector j; pools across all destination sectors

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- Does having more coworkers coming from a certain industry makes job switchers more likely to move to "better quality" firms within a sector?
- Sample includes all individuals that transition into sector j; pools across all destination sectors
- Two outcome variables as proxies for job qualities: AKM firm FE at the new job and the tenure at new job

Firm choice for switchers

	Firm Wage FE	Firm Wage FE	Tenure at New Job	Tenure at New Job
Current-year Coworker	0.133***	0.139***	0.458***	0.471***
	(0.0110)	(0.0113)	(0.0153)	(0.0104)
Non-current-year Coworker	0.723***	0.713***	0.531***	0.545***
	(0.0197)	(0.0193)	(0.0094)	(0.0104)
AKM Person FE		✓		✓
Mean of Dep. Var	12.4	12.4	1.92	1.92
R^2	0.39	0.40	0.121	0.13
Observations	8,743,260	7,882,124	11,010,158	8,236,194

- Higher share of coworkers in sector $\Rightarrow$ AKM firm FE $\uparrow$ (0.67 p.p in the firm wage distribution), tenure $\uparrow$ (5.5 months)

back

Productivity Shock

Sectors	Shock
Agriculture and Mining	4.1%
Manufacturing	-0.9%
Utilities, Water Supply, and Waste Management	3.8%
Construction	0.8%
Trade, Transportation, and Storage	-0.8%
Information and Professional Services	2.1%
Financial, Insurance, and Real Estate	-0.7%
Accommodation, Administration, Arts, Other Services	-4.8%
Education and Health Services	-4.2%
All Private Sectors	-0.3%

back

Job-switching elasticity

$$(\rho-1)\log\frac{\mu_t^{jn,rn|n}}{\mu_t^{jn,jn|n}}+\log\overbrace{\frac{\mu_{t-1}^{jn,rn|n}}{\mu_{t-1}^{jn,jn|n}}}^{\text{continuation value}}+\frac{\phi}{\zeta}\log\overbrace{\frac{\mu_t^{rn,n}}{\mu_t^{jn,n}}}^{\text{sector stayers}}+\phi\log\overbrace{\frac{\mu_t^{rn,rn|n}}{\mu_t^{jn,jn|n}}}^{\text{firm stayers}}+\mathcal{Z}_t=$$

$$\frac{1}{\nu\zeta}\left[\log\frac{w_t^{rn}}{w_t^{jn}}-\frac{\rho}{\rho+\phi}\log B^{rn}\right]-\frac{\phi\beta_0(l_t^{n,rn}-l_{t-1}^{n,jn})}{(\rho+\phi)\nu\zeta}\left(\bar{z}_t-\log(Z_t^n)\right)+\epsilon_t$$

- Coworker influence:
  $\mathcal{Z}_t=\frac{\beta_1(l_t^{n,jn}-l_{t-1}^{n,jn})}{(\rho+\phi)\nu\zeta}\log B^{rn}+\frac{\alpha_1}{\nu\zeta}(l_t^{rn,jn}-l_{t-1}^{rn,jn})\bar{\kappa}-\frac{\rho\beta_1 l_t^{n,jn}}{(\rho+\phi)\nu\zeta}\log B^{rn}+\frac{\rho}{\nu\zeta}\bar{\kappa}-\frac{\rho\alpha_1}{\nu\zeta}l_t^{rn,jn}\bar{\kappa}$

back

Sector-Level Network and Transition Costs

$$\log \frac{\mu_t^{jn,k}}{\mu_t^{jn,n}} + \log \frac{\mu_t^{ik,n}}{\mu_t^{ik,k}} = -\zeta(S_t^{jn,k} - S_t^{ik,k}) - \zeta(S_t^{jn,n} - S_t^{ik,n})$$

$$+ \frac{\beta_0}{\nu(\rho+\phi)}\left[(I_t^{k,jn} - I_t^{k,ik})\log(Z_t^k) - (I_t^{n,jn} - I_t^{n,ik})\log(Z_t^n)\right] + \frac{\alpha_0}{\nu}\left[I^{k,jn} + I^{n,ik}\right]\bar{\kappa}^{n,k}$$

- $(I_t^{k,jn} - I_t^{k,ik})\log(Z_t^k) - (I_t^{n,jn} - I_t^{n,ik})\log(Z_t^n)$: sum of sector-level coworker influence on TFP
- $\left[I^{k,jn} + I^{n,ik}\right]\bar{\kappa}^{n,k}$: sum of sector-level coworker influence on transition costs
- $\bar{\kappa}^{n,k}$ varies by n, k

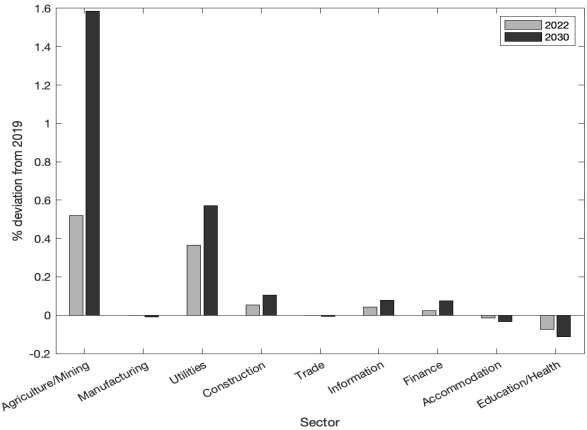
back

Parameter Values

Parameter	Value	Statistic	Related Moment
ρ	0.05	Value	Discount rate
ϕ	2.3	Value	Job-Switching Rate
ζ	0.2	Value	Correlation of taste shocks within each sector
$\bar{\kappa}$	15.0	Value	Levels of cross-firm flows by firm pair
γ	0.37	Value	Labor Share of Income
α^n	[0.01, 0.29]	Range	Consumption share of sectoral goods
η	0.14	Value	Autocorrelation of sector-specific productivity process
$\log(B)^{jn}$	[-1.50, 1.37]	Range	Mean of firm-specific wage and production
$\log(Z^n)$	[3.10, 4.33]	Range	Level of sector-specific productivity process
δ^n	[0.03, 0.08]	Range	Variance of sector-specific productivity process
ν	2.63	Value	Elasticities of cross-firm flows w.r.t. wage
$\bar{\kappa}^{n,k}$	[1.28, 12.36]	Range	Levels of cross-sector flows by sector pair
α_0	0.06	Value	Levels of cross-sector flows shared across all sectors
α_1	7.05	Value	Levels of cross-firm flows
β_0	4.17	Value	Elasticities of cross-sector flows w.r.t. TFP
β_1	1.16	Value	Elasticities of cross-firm flows w.r.t. TFP

back

More on labor reallocation



- Labor keeps flowing towards sectors that are positively shocked, years after the shock takes place

back

Decomposing the sources of welfare changes

Table: Welfare Impact of productivity shock (% deviation from s.s.)

	Mechanism			
	Productivity	Reallocation	Wage	Total
Coworker Network (2022)	-0.0042	-0.0023	0.1827	0.1762
Coworker Network (2030)	-0.0004	-0.0055	0.3241	0.3182
No Network (2022)	-0.0040	0.0007	0.0005	-0.0028
No Network (2030)	-0.0004	0.0003	0.0009	0.0007

- Maintaining worker-employer relationships is more beneficial according to the model with coworker networks
- Mostly due to the GE effect of changes in wages

back